Multi-Agent Requirement Clarification via Missing Information Detection

Anonymous Author(s)

Abstract

Natural language requirements often omit essential information and contain ambiguity during elicitation. This paper studies multi-agent requirement clarification via missing information detection. The proposed research framework coordinates interviewer, end-user, and analyst agents to detect missing functional, constraint, role, data, and exception information, then uses knowledge-guided clarification templates to ask targeted follow-up questions. The study is scoped to G1-elicitation and evaluates whether this workflow improves requirement completeness, missing information coverage, clarification effectiveness, and human effort.

Keywords

requirement elicitation, requirement clarification, missing information detection, multi-agent collaboration, knowledge-guided analysis, human-in-the-loop

1 Introduction

1.1 Background and Motivation

Requirement elicitation determines whether stakeholder goals, domain constraints, and operational assumptions are captured before requirements are formalized. In intelligent requirements engineering, requirements increasingly become the interface through which stakeholders communicate intent to AI-supported tools [2]. However, initial natural-language requirements are frequently partial, informal, and context-dependent.

1.2 Challenges of Requirement Clarification

LLM-based elicitation can generate fluent requirement statements, but fluency does not guarantee completeness or clarity. Initial stakeholder inputs may omit actors, data objects, exception cases, quality constraints, or business rules. Static requirement generation is especially limited because it may rewrite vague input instead of asking targeted follow-up questions. Prior work on controlled natural language also shows that reducing ambiguity while preserving stakeholder readability remains challenging [1].

1.3 Multi-Agent Requirement Clarification

This paper focuses on multi-agent requirement clarification via missing information detection. The central idea is to use specialized agents to identify incomplete requirement elements, generate dynamic clarification questions, and keep human stakeholders responsible for approving elicitation decisions. Knowledge-assisted analysis guides the agents toward functional, constraint, role, data, and exception gaps.

1.4 Contributions

The paper is organized around the following contributions:

- A research-problem-driven workflow for multi-agent requirement clarification in G1-elicitation.
- A missing-information detection scheme covering functional, constraint, role, data, and exception information.
- A knowledge-assisted clarification method that uses requirement knowledge, templates, and prompt strategies for dynamic follow-up.
- An experimental design for evaluating requirement completeness, missing-information coverage, clarification effectiveness, and human effort.

2 Related Work

2.1 Traditional Requirement Elicitation

Traditional requirement elicitation relies on interviews, workshops, scenarios, questionnaires, and document analysis to externalize stakeholder needs. These techniques provide rich human context but depend heavily on analyst expertise and may miss implicit domain assumptions.

2.2 Requirement Clarification

Requirement clarification addresses incomplete, vague, or ambiguous stakeholder statements through follow-up questions and iterative refinement. This paper treats clarification as a focused elicitation activity rather than a downstream specification or implementation task.

2.3 LLM-Based Requirement Elicitation

LLMs can support elicitation by summarizing stakeholder input, proposing requirement candidates, and generating follow-up questions. The key limitation is that generated questions may be generic, weakly connected to missing information, or unsupported by explicit requirement knowledge.

2.4 Multi-Agent Collaboration

Multi-agent collaboration assigns elicitation responsibilities to specialized roles. In this paper, collaboration is used narrowly for requirement clarification: one agent conducts the interview, one simulates or represents end-user perspectives, and one analyzes requirement quality during the dialogue.

2.5 Human-in-the-Loop Requirement Engineering

Human-in-the-loop requirement engineering keeps stakeholders and analysts responsible for validating questions, answers, and clarified requirement statements. This line of work motivates interactive clarification workflows in which AI agents assist elicitation without replacing stakeholder judgment.

3 Methodology

3.1 Framework Overview

The framework takes an incomplete natural-language requirement as input and produces a clarified requirement statement with its elicitation record. It follows a lightweight loop: detect missing information, assign clarification responsibilities to agents, ask knowledge-guided follow-up questions, and let humans validate the refined requirement. The scope is limited to requirement elicitation.

3.2 Multi-Agent Requirement Clarification Workflow

The clarification workflow uses three complementary agents.

3.2.1 Interviewer Agent. The interviewer agent manages the elicitation dialogue. It asks concise follow-up questions, adapts the question order based on stakeholder responses, and avoids unrelated prompts.

3.2.2 End-User Agent. The end-user agent represents usage perspectives during clarification. It helps expose user goals, operational contexts, expected interactions, and practical constraints that may be absent from the original requirement.

3.2.3 Analyst Agent. The analyst agent reviews the evolving requirement statement. It checks whether the current version contains sufficient actor, action, object, condition, constraint, data, and exception information for elicitation.

3.3 Missing Information Detection

Missing information detection identifies which requirement elements must be clarified before the requirement can be considered complete for the elicitation task.

3.3.1 Functional Information. Functional information covers actors, goals, actions, inputs, outputs, triggers, and expected system responses.

3.3.2 Constraint Information. Constraint information covers business rules, quality constraints, timing limits, access restrictions, and usage conditions.

3.3.3 Role, Data, and Exception Information. Role, data, and exception information is treated as high priority because these gaps often determine whether stakeholders interpret a requirement consistently.

3.4 Knowledge-Assisted Clarification

Knowledge assists the agents in selecting what to ask and how to ask it.

3.4.1 Requirement Knowledge. Requirement knowledge includes domain terms, stakeholder roles, requirement slots, constraint categories, and elicitation checklists.

3.4.2 Clarification Templates. Clarification templates map missing information types to targeted follow-up questions. For example, a missing actor triggers a role question, while a missing exception triggers an alternative-flow question.

3.4.3 Prompt Strategies. Prompt strategies constrain agents to ask one focused question at a time, justify the missing information type, and avoid introducing requirements not supported by stakeholder input.

3.5 Human-in-the-Loop Interaction

Human participants validate clarification questions, provide answers, and approve revised requirement statements. The workflow preserves human control over requirement meaning while using agents to reduce missed clarification opportunities.

3.6 Iterative Requirement Refinement

The workflow repeats until the remaining missing information is resolved or explicitly marked as unknown by the stakeholder. Each iteration updates the current requirement statement and records the question-answer pair that motivated the refinement.

4 Experiments

4.1 Research Questions

The empirical study evaluates whether the clarification workflow improves elicitation outcomes.

RQ1: Does multi-agent clarification improve requirement completeness compared with non-iterative elicitation?
RQ2: How accurately does missing information detection identify functional, constraint, role, data, and exception gaps?
RQ3: How much do knowledge assistance and agent collaboration contribute to clarification effectiveness?
RQ4: Does the workflow reduce human effort while preserving stakeholder control?

4.2 Experimental Setup

The setup compares elicitation sessions that start from the same incomplete natural-language requirements. Each session produces clarification questions, stakeholder answers, refined requirements, and an elicitation log for evaluation.

4.3 Datasets

Datasets should consist of requirement elicitation scenarios with incomplete initial descriptions and reference missing-information annotations. The dataset description should report domain coverage, scenario size, and annotation procedure without introducing results.

4.4 Baselines

Baselines should remain within requirement elicitation. Candidate baselines include single-agent LLM clarification, prompt-only clarification, checklist-based elicitation, and analyst-authored clarification questions.

4.5 Evaluation Metrics

4.5.1 Requirement Completeness. Requirement completeness measures the proportion of required elicitation elements captured after clarification.

4.5.2 Missing Information Coverage. Missing information coverage measures how many annotated missing elements are detected and addressed by clarification questions.

4.5.3 Clarification Effectiveness. Clarification effectiveness measures whether generated questions lead to useful stakeholder answers and improved requirement statements.

4.5.4 Human Effort Reduction. Human effort reduction measures the number of clarification turns, manual edits, or reviewer interventions needed to reach an acceptable requirement.

4.6 Main Results

This subsection should report aggregate comparisons after experiments are completed. No numerical claims or result tables are included before empirical execution.

4.7 Ablation Study

The ablation study should remove or vary major workflow components, including missing-information detection, requirement knowledge, agent collaboration, and human validation.

4.8 Case Study

The case study should present one requirement elicitation scenario, the detected missing information, the generated clarification questions, and the final clarified requirement statements.

5 Threats to Validity

5.1 Internal Validity

Internal validity threats may arise from differences in stakeholder responses, analyst judgment, or clarification ordering across sessions. The study should control initial requirements and use consistent evaluation procedures.

5.2 External Validity

External validity threats arise from limited domains, scenario complexity, and participant diversity. The results should be interpreted as evidence for requirement elicitation scenarios similar to those evaluated.

5.3 Construct Validity

Construct validity threats concern whether the selected metrics adequately capture requirement completeness, missing-information coverage, clarification effectiveness, and human effort. The study should define each metric before annotation and use clear coding guidelines.

6 Conclusion

This paper studies multi-agent requirement clarification through missing information detection. The structure focuses on knowledge guided elicitation, dynamic follow-up questions, and human validation. The scope remains G1 elicitation and targets requirement completeness and clarification effectiveness.

References

[1] Ikram Darif, Ghizlane El Boussaidi, Segla Kpodjedo, and Cristiano Politowski. 2026. Controlled Natural Language for Requirements Specification: A Systematic Literature Review. *ACM Comput. Surv.* 58, 7 (Jan. 2026), 175:1–175:36. doi:10.1145/3778169

[2] Diana Robinson, Christian Cabrera, Andrew D. Gordon, Neil D. Lawrence, and Lars Mennen. 2025. Requirements Are All You Need: The Final Frontier for End-User Software Engineering. *ACM Trans. Softw. Eng. Methodol.* 34, 5 (May 2025), 141:1–141:22. doi:10.1145/3708524

A Appendix

The appendix may include elicitation scenario templates, missing-information annotation guidelines, example clarification logs, and supplementary evaluation rubrics. It should not introduce claims or artifacts outside the requirement elicitation scope.